Consider the one-dimensional potential

$$V(x) = \begin{cases} +\infty, & |x| > a, \\ 0, & b < |x| < a, \\ V_0, & |x| < b, \end{cases} \qquad a > b > 0.$$

The wavefunction vanishes at the infinite walls, $\psi(\pm a) = 0$.

1 I

1.1 (1)

When $V_0 = 0$, the time-independent Schrödinger equation inside the well is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi, \qquad \frac{d^2\psi}{dx^2} = -k^2\psi, \qquad k = \frac{\sqrt{2mE}}{\hbar}.$$

Since the potential is symmetric, the eigenfunctions can be chosen to have definite parity.

For an even-parity state, assume

$$\psi(x) = A\cos(kx).$$

The boundary condition $\psi(a) = 0$ requires

$$\cos(ka) = 0 \quad \Longrightarrow \quad ka = n_{\text{odd}}\frac{\pi}{2}, \qquad n_{\text{odd}} = 1, 3, 5, \ldots.$$

The ground state therefore corresponds to $k = \pi/(2a)$. After normalization,

$$\boxed{\psi_1(x) = \frac{1}{\sqrt{a}}\cos\left(\frac{\pi x}{2a}\right)}, \qquad \boxed{E_1 = \frac{\hbar^2\pi^2}{8ma^2}} \qquad (|x| < a).$$

Outside the well, $\psi_1(x) = 0$.

1.2 (2)

For the general eigenstates, we consider even and odd parity separately.

Even parity:

$$\psi_n(x) = A\cos(k_n x), \qquad k_n a = n\frac{\pi}{2}, \qquad n = 1, 3, 5, \ldots.$$

Odd parity:

$$\psi_n(x) = B\sin(k_n x), \qquad k_n a = n\frac{\pi}{2}, \qquad n = 2, 4, 6, \ldots.$$

For both cases, normalization gives

$$1 = |A|^2\int_{-a}^{a}\cos^2(k_n x)\,dx = |B|^2\int_{-a}^{a}\sin^2(k_n x)\,dx = a|A|^2 = a|B|^2,$$

where the corresponding integral is used in each parity sector. Choosing real positive amplitudes, $A = B = 1/\sqrt{a}$. Thus,

$$\boxed{E_n = \frac{\hbar^2 n^2\pi^2}{8ma^2}}, \qquad n = 1, 2, 3, \ldots,$$

and, for $|x| < a$,

$$\boxed{\psi_n(x) = \begin{cases} \frac{1}{\sqrt{a}} \cos\left(\frac{n\pi x}{2a}\right), & n \text{ odd}, \\ \frac{1}{\sqrt{a}} \sin\left(\frac{n\pi x}{2a}\right), & n \text{ even}. \end{cases}}$$

For $|x| \geq a$, $\psi_n(x) = 0$.

2 II

When $V_0 > E$, the Schrödinger equation in the central region $|x| < b$ is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V_0\psi = E\psi,$$

so that

$$\frac{d^2\psi}{dx^2} = \rho^2\psi, \qquad \rho = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

In the two outer regions, $b < |x| < a$, we still have $\psi'' = -k^2\psi$, with $k = \sqrt{2mE}/\hbar$.

2.1 (3)

For an even potential, $V(x) = V(-x)$, the Hamiltonian commutes with the parity operator:

$$[\hat{P}, \hat{H}] = 0.$$

Therefore, if $\psi(x)$ is an energy eigenfunction, $\psi(-x)$ is also an eigenfunction with the same energy.

To verify the absence of degeneracy in one dimension, consider two eigenfunctions ψ_1 and ψ_2 with the same energy E. Their Wronskian is

$$W(x) = \begin{vmatrix} \psi_1(x) & \psi_2(x) \\ \psi_1'(x) & \psi_2'(x) \end{vmatrix} = \psi_1\psi_2' - \psi_2\psi_1'.$$

Differentiating gives

$$\frac{dW}{dx} = \psi_1\psi_2'' - \psi_2\psi_1''.$$

Using the time-independent Schrödinger equation

$$\psi_i'' = \frac{2m}{\hbar^2}[V(x) - E]\psi_i,$$

we obtain

$$\frac{dW}{dx} = \frac{2m}{\hbar^2}[V(x) - E](\psi_1\psi_2 - \psi_2\psi_1) = 0.$$

Hence W is constant within the well. Since both eigenfunctions satisfy $\psi_i(a) = 0$, we have $W(a^-) = 0$, and thus $W = 0$ everywhere (the wavefunctions and their first derivatives are continuous at $x = \pm b$). Consequently, the two solutions are linearly dependent, and the energy level is nondegenerate.

It follows that

$$\psi(-x) = c\psi(x).$$

Applying parity twice,

$$\psi(x) = c\psi(-x) = c^2\psi(x), \qquad c^2 = 1.$$

Therefore,

$$\boxed{\psi(-x) = \pm\psi(x)},$$

so each eigenfunction has definite parity.

2.2 (4)

For an even-parity state, consider the two outer regions $-a < x < -b$ and $b < x < a$. Since

$$\frac{d^2\psi}{dx^2} = -k^2\psi,$$

and $\psi(\pm a) = 0$, we may write

$$\psi(x) = \begin{cases} B\sin[k(x+a)], & -a < x < -b, \\ A\sin[k(x-a)], & b < x < a. \end{cases}$$

The even-parity condition $\psi(-x) = \psi(x)$ gives $B = -A$. Thus,

$$\boxed{\psi(x) = \begin{cases} -A\sin[k(x+a)], & -a < x < -b, \\ A\sin[k(x-a)], & b < x < a. \end{cases}}$$

2.3 (5)

In the central region $|x| < b$,

$$\frac{d^2\psi}{dx^2} = \rho^2\psi.$$

The general solution is a linear combination of $\cosh(\rho x)$ and $\sinh(\rho x)$. For even parity, only the even term remains:

$$\boxed{\psi(x) = C\cosh(\rho x)}, \qquad |x| < b.$$

2.4 (6)

Requiring continuity of ψ and $d\psi/dx$ at $x = b$, we obtain

$$\begin{aligned} A\sin[k(b-a)] &= C\cosh(\rho b), \\ kA\cos[k(b-a)] &= \rho C\sinh(\rho b). \end{aligned}$$

Dividing the first equation by the second gives

$$\frac{1}{k}\tan[k(b-a)] = \frac{1}{\rho\tanh(\rho b)}.$$

Since $\tan[k(b-a)] = -\tan[k(a-b)]$, the even-parity energy eigenvalues satisfy

$$\boxed{\frac{1}{k}\tan[k(a-b)] = -\frac{1}{\rho\tanh(\rho b)}}.$$

2.5 (7)

For an odd-parity state, the outer-region solutions have $B = A$, while the central-region solution is odd:

$$\psi(x) = \begin{cases} A\sin[k(x+a)], & -a < x < -b, \\ D\sinh(\rho x), & |x| < b, \\ A\sin[k(x-a)], & b < x < a. \end{cases}$$

Applying continuity at $x = b$,

$$A\sin[k(b-a)] = D\sinh(\rho b),$$
$$kA\cos[k(b-a)] = \rho D\cosh(\rho b).$$

Dividing the equations,

$$\frac{1}{k}\tan[k(b-a)] = \frac{1}{\rho}\tanh(\rho b).$$

Therefore, the odd-parity energy eigenvalues satisfy

$$\boxed{\frac{1}{k}\tan[k(a-b)] = -\frac{1}{\rho}\tanh(\rho b)}.$$

2.6 (8)

In the limit $V_0 \to +\infty$, we have $\rho \to +\infty$, and

$$\tanh(\rho b) \to 1.$$

For both the even- and odd-parity eigenvalue equations, the right-hand side tends to zero. Thus,

$$\tan[k(a-b)] = 0, \qquad k(a-b) = n\pi, \qquad n = 1, 2, 3, \ldots.$$

The lowest allowed wave number is $k = \pi/(a-b)$, giving

$$\boxed{E_{\text{even,min}} = E_{\text{odd,min}} = \frac{\hbar^2\pi^2}{2m(a-b)^2}}.$$

Hence the lowest even- and odd-parity states are degenerate.

Physically, when the central barrier becomes infinitely high, the two wells $(-a, -b)$ and (b, a) are completely disconnected. Each has width $a - b$, and a state localized in either well has the same energy. Their symmetric and antisymmetric combinations give the degenerate even- and odd-parity states. For finite V_0, tunneling through the central barrier couples the two wells and lifts this degeneracy.